METRIC

MIL-DTL-181C(OS)
23 June 2023
SUPERSEDING
MIL-DTL-181B(OS)
5 January 2021

DETAIL SPECIFICATION

# WAX, CANDELILLA

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one grade of candelilla wax.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E1 - Standard Specification for ASTM Liquid-in-Glass Thermometers

ASTM E11 - Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves

(Copies of these documents are available online at www.astm.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Material and workmanship. The wax shall be a naturally-occurring extract, clear, and free from foreign matter.

3.2 Chemical and physical properties. Candelilla wax shall conform to the chemical and physical properties requirements specified in table I.

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AMSC N/A FSC 9160

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TABLE I. Chemical and physical properties.

| Property | Minimum | Maximum | Notes |
|---|---|---|---|
| Volatile matter, % | -- | 1.5 | -- |
| Solidification point, °C | 63 | 70 | -- |
| Percent insoluble in carbon tetrachloride | -- | 1 | -- |
| Acid number | -- | 22 | -- |
| Saponification number | 45 | 65 | -- |
| Mineral acidity | N/A | N/A | Shall not be acidic (see 4.5.6) |
| Ash, % | -- | 0.5 | -- |

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5 as specified in table II.

TABLE II. Requirements/verification cross reference matrix.

| Method of Verification | | | | Classes of Verification | |
|---|---|---|---|---|---|
| 1—Analysis | 3—Examination | | | A—First article | |
| 2—Demonstration | 4—Test | | | B—Conformance | |

| Requirement | Description | Verification Methods | | | | Verification Class | | Verification |
|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | A | B | |
| 3.1 | Material and workmanship | | | X | | | X | 4.4 |
| 3.2 | Volatile matter, % | X | | | X | | X | 4.5.1 |
| 3.2 | Solidification point, °C | X | | | X | | X | 4.5.2 |
| 3.2 | Percent insoluble in carbon tetrachloride | X | | | X | | X | 4.5.3 |
| 3.2 | Acid number | X | | | X | | X | 4.5.4 |
| 3.2 | Saponification number | X | | | X | | X | 4.5.5 |
| 3.2 | Mineral acidity | X | | | X | | X | 4.5.6 |
| 3.2 | Ash, % | X | | | X | | X | 4.5.7 |

4.2 Size of lots. Unless otherwise specified (see 6.2), the maximum lot size shall be 4,536 kilograms (10,000 pounds).



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4.3 Sampling. A minimum of 10 percent of the containers in the lot shall be selected in such a manner as to be representative of the lot. When lots comprise less than 100 containers, either 10 containers or all containers in the lot shall be selected. Approximately 227 grams (½ pound) of the material shall be removed from each selected container. Each primary sample shall be labeled so that the container from which it was taken can be identified. The individual samples shall be pulverized until the material passes through a 4.76-millimeter (U.S. Standard No. 4) sieve. An approximate 227-gram (½-pound) composite sample shall be made of equal parts of each primary sample. All samples shall be placed in airtight containers and labeled to show the name of the material, manufacturer, contract or order number, lot number, and the number of pounds in the lot. The primary samples shall be held for possible future examination should the tests of the composite sample show that the lot fails to meet the requirements herein. All acceptance tests shall be made on the composite sample representative of the test.

4.4 Examination of product. The wax shall be visually examined for conformance to 3.1.

4.5 Test methods. The following test methods shall be used to determine conformance to 3.2.

4.5.1 Volatile matter. Volatile matter shall be determined as follows:

a. Weigh approximately 5 grams of the sample in a tared glass-stoppered weighing bottle.

b. Heat the sample in an oven for 5 hours at 100 to 105 °C, cool in a desiccator, and weigh.

c. Calculate the loss of weight as percentage of volatile matter using the following equation:

$$\%\ \text{Volatile matter} = \frac{(A - B)}{W} \times 100$$

Where:

A = Weight of bottle, stopper, and sample before heating
B = Weight of bottle, stopper, and sample after heating
W = Weight of sample

4.5.2 Solidification point. The solidification point shall be determined as follows:

a. Break up 40 to 50 grams of the sample in a mortar and pestle to a size that is fine enough to pass through a standard 1.7-millimeter (U.S. Standard No. 12) sieve conforming to ASTM E11.

b. Dry the sample for 4 hours at approximately 40 °C.

c. Place the dry sample into the inner tube of the solidification-point apparatus shown on figure 1.

d. Place the tube into a 95 to 100 °C glycerin bath and allow the sample to melt.

e. Refill the inner tube with additional wax as necessary to get the liquid to the level such that the thermometer conforming to ASTM E1, type 93C (76-millimeter partial immersion, 60 to 90 °C range, 0.1 °C increments) is at the proper immersion level, the bulb is in the center of the liquid wax, and the stirrer can sufficiently agitate the solution.

f. Stopper the test tube with the two-hole stopper, fitted with the stirring rod with a loop end, and the thermometer.

g. Remove the tube from the bath, wipe the outside of it clean, and place the test tube into the solidification-point apparatus.

h. Stir the molten material constantly (manually or by automated stirrer) while continually monitoring the temperature of the sample. The temperature will rise as solidification begins.

i. Record the temperature every 15 seconds until the maximum reading is obtained, which is the solidification point.



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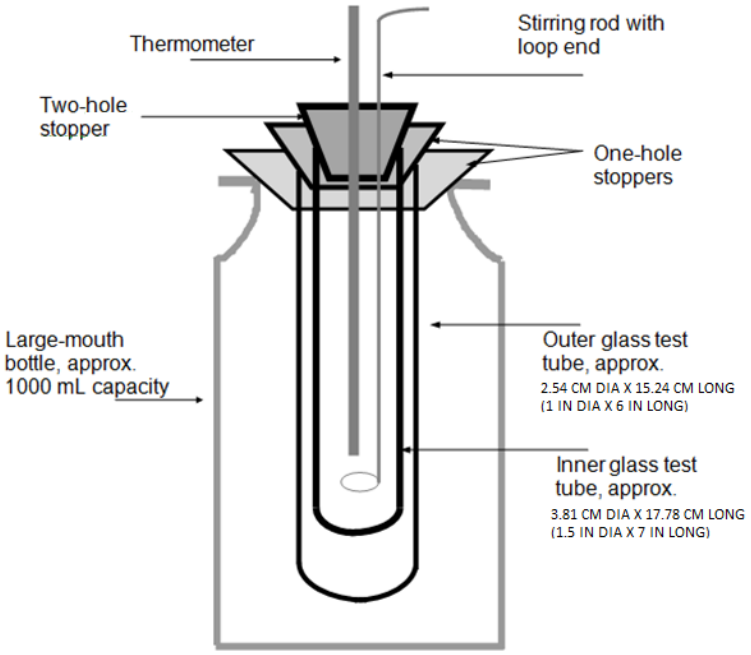



FIGURE 1. Solidification-point apparatus.

4.5.3 Material insoluble in carbon tetrachloride. The material insoluble in carbon tetrachloride shall be determined as follows:

a. Dissolve a weighed portion of approximately 5 grams of the finely-divided sample in 100 milliliters of hot carbon tetrachloride.

b. Filter the solution through a tared glass filtering crucible, medium porosity.

c. Wash the crucible and residue thoroughly with two 25-milliliter portions of hot carbon tetrachloride.

d. Dry the crucible for 1 hour at 100 °C, cool in a desiccator, and weigh.

e. Calculate the gain in weight to percentage carbon tetrachloride insoluble matter using the following equation:

$$\%\ \text{material insoluble in carbon tetrachloride} = \frac{(A - B)}{W} \times 100$$

Where:

A = Weight of crucible and residue after drying
B = Weight of empty crucible
W = Weight of sample

4.5.3.1 Alternate method: material insoluble in toluene. The material insoluable in toluene shall be determined as follows:

a. Dissolve a weighed portion of approximately 5 grams of the finely-divided sample in 100 milliliters of toluene.

b. Filter the solution through a tared glass filtering crucible, medium porosity.

c. Wash the crucible and residue thoroughly with two 25-milliliter portions of toluene.



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d. Dry the crucible for 1 hour at 95 to 105 °C, cool in a desiccator, and weigh.

e. Calculate the gain in weight to percentage toluene insoluble matter using the following equation:

$$\%\ \text{material insoluble in toluene} = \frac{(A - B)}{W} \times 100$$

Where:

A = Weight of crucible and residue after drying
B = Weight of empty crucible
W = Weight of sample

4.5.4 Acid number. The acid number shall be determined as follows:

a. Prepare an ethanol-toluene solution (2 parts 95 percent ethanol or USSD3A-denatured ethanol to 1 part toluene) in sufficient quantity to have 100 milliliters for each test.

b. Add several drops of 1 percent phenolphthalein in the ethanol indicator solution to the mixture and titrate to the first persistent pink endpoint using 0.05 normal (N) sodium hydroxide or potassium hydroxide.

c. Transfer approximately 2 grams of the sample, weighed to the nearest 0.001 gram, into a 250-milliliter acid value flask and add 100 milliliters of ethanol-toluene solution.

d. Carefully warm the flask on a hot plate or water bath to completely dissolve the wax.

e. Add 3 to 5 drops of the phenolphthalein indicator solution and quickly titrate the hot solution to the first persistent pink endpoint using the 0.05N sodium hydroxide (or potassium hydroxide), swirling the flask vigorously during the titration.

f. If the wax precipitates during titration, reheat the sample to redissolve the wax, then continue the titration.

g. Calculate the acid number using the following equation:

$$\text{Acid number} = \frac{VN \times 56.1}{W}$$

Where:

V = Milliliters of sodium hydroxide (or potassium hydroxide) solution used
N = Normality of sodium hydroxide (or potassium hydroxide) solution
W = Weight of sample

4.5.5 Saponification number. The saponification number shall be determined as follows:

a. Transfer approximately 1 gram of the pulverized sample, weighed to the nearest 0.001 gram, into a 250-milliliter Erlenmeyer flask.

b. Measure 50 milliliters of 0.1N alcoholic potassium hydroxide solution into the flask containing the test sample.

c. Add an equal amount of the potassium hydroxide solution to a 250-milliliter Erlenmeyer flask containing no sample for use as a blank.

d. Connect both flasks to condensers and reflux on a hot plate for 3 hours.

e. Remove the flasks from the condensers, add 3 to 5 drops of phenolphthalein (1 gram in 100 milliliters of 95 percent ethanol), and titrate both the sample and the blank with 0.1N hydrochloric acid until the pink color disappears and does not return for at least 10 seconds.

f. If the solution congeals on cooling, or if the soap separates on cooling, conduct the titration as near boiling as is feasible.

g. Frequently swirl the contents of the flasks during titration.

h. Calculate the saponification number using the following equation:



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$$\text{Saponification number} = \frac{(V - v)N \times 56.1}{W}$$

Where:

V = Milliliters of acid required for blank
v = Milliliters of acid required for sample
N = Normality of acid used
W = Weight of sample

4.5.6 Mineral acidity. The mineral acidity shall be determined as follows:

a. Place 5 grams of the finely-divided sample in a 250-milliliter beaker, add 100 milliliters of cold, distilled water, shake thoroughly, and filter through filter paper into a flask.

b. Wash the beaker and sample with a small quantity of distilled water and filter into the flask containing the first filtrate.

c. Add 5 to 10 drops of methyl-orange indicator solution to the filtrate and observe whether the solution gives an acidic reaction. If the solution appears red, it shall be considered acidic; if the solution appears yellow, it shall not be considered acidic.

4.5.7 Ash. The ash percentage shall be determined as follows:

a. Heat a weighed sample, accurate to the nearest 0.01 gram, of approximately 5 to 10 grams in a previously ignited and weighed evaporating dish (if not platinum, porcelain is acceptable) over a free flame until the vapors ignite.

b. Avoid loss by spattering until all volatile, combustible matter has been removed and the sample is charred.

c. Transfer the dish and contents to a muffle furnace and ignite to constant weight at 600 °C.

d. Allow to cool in a desiccator and weigh.

e. Calculate the increase in weight as percentage of ash in the sample using the following equation:

$$\%\ \text{Ash} = \frac{(A - B)}{W} \times 100$$

Where:

A = Weight of dish and residue after heating
B = Weight of dish
W = Weight of sample

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Candelilla wax is intended for use in double-base propellants such as AA-2, AA-6, N-5, M37, and M44, which are used in rockets and other military propulsion systems. Existing stock of candelilla wax manufactured to MIL-W-181A(OS), dated 18 August 1975, and MIL-DTL-181B(OS), dated 5 January 2021, is acceptable for use until depleted.



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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Maximum size of lot if other than specified (see 4.2).

c. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

AA-2 propellant
AA-6 propellant
M37 propellant
M44 propellant
N-5 propellant

6.4 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 9160-2023-001)

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